Flicker Noise in Silicon Bipolar Transistors: Physical Origin, Compact Modeling, and Circuit-Level Impact

Abstract:

This paper reviews flicker noise in silicon bipolar transistors. Flicker noise, also called 1/f noise, is an important low-frequency noise source in bipolar junction transistors. It is usually observed as a noise spectral density that increases when frequency decreases. In silicon BJTs, flicker noise is strongly related to base current fluctuation, carrier trapping and detrapping, trap-assisted recombination, and surface or interface defects. In practical circuit design, flicker noise is important because it limits the performance of low-frequency amplifiers, bandgap references, precision analog circuits, and radio-frequency oscillators. This paper first introduces the main noise sources in bipolar transistors, including thermal noise, shot noise, and flicker noise. Then it explains the physical origin of flicker noise using simple device-level concepts. After that, compact modeling methods and simulation program with integrated circuit emphasis noise parameters are discussed. The dependence of flicker noise on bias current, device geometry, and fabrication process is also reviewed. Finally, the paper summarizes the circuit-level impact of BJT flicker noise and discusses possible noise reduction techniques.

1. Introduction:

Silicon bipolar junction transistors, usually called BJTs, are widely used in analog, mixed-signal, precision, and radio-frequency circuits. Although CMOS technology is dominant in many digital systems, BJTs are still important because they can provide high transconductance, good matching, strong current-driving capability, and useful analog characteristics.

However, every real transistor produces noise. Noise is an unwanted random fluctuation of voltage or current. In a circuit, noise can reduce signal quality, limit resolution, and degrade accuracy. For example, in a low-frequency amplifier, transistor noise may hide a weak input signal. In a bandgap reference, noise can create small but important voltage fluctuations. In an oscillator, low-frequency transistor noise can be converted into phase noise around the carrier signal.

Among different noise mechanisms, flicker noise is especially important at low frequency. Flicker noise is also called 1/f noise because its spectral density is often approximately proportional to $1/f$. A simple expression is: $S(f) \propto 1/f^{\gamma}$, where $S(f)$ is the noise spectral density, f is frequency, and γ is usually close to 1.

In silicon BJTs, flicker noise is often associated with base current fluctuation. These fluctuations are usually connected to defects, traps, recombination centers, and imperfect surfaces or interfaces inside the transistor. Studies on silicon bipolar transistors have shown that the input-referred flicker noise can often be modeled as a base current noise source and that the noise level can depend strongly on base current and emitter area (Deen et al., 1999; Simoen et al., 1998).

This paper aims to systematically elaborate the physical characteristics of flicker noise in bipolar junction transistors (BJTs) in a clear and accessible manner. The overall discussion centers on three core aspects: the dominant noise sources within BJT devices, the fundamental physical mechanisms responsible for the generation of flicker noise, and the practical impacts of flicker noise on circuit performance in real engineering applications.

2. Noise Sources in Bipolar Transistors

Several kinds of noise exist in bipolar transistors. The most common ones are thermal noise, shot noise, and flicker noise. To understand flicker noise clearly, it is useful to first compare it with the other two basic noise sources.

2.1. Thermal Noise

Thermal noise is caused by the random thermal motion of charge carriers. It exists in resistive materials even when no DC current flows. Because temperature causes carriers to move randomly, a resistor produces a random voltage or current noise.

For a resistor R, the voltage noise spectral density is commonly written as: $S_v = 4kTR$, where k is Boltzmann's constant, T is absolute temperature, and R is resistance.

Thermal noise is often called white noise because its spectral density is approximately constant over a wide frequency range. In BJT circuits, thermal noise can come from base resistance, emitter resistance, collector resistance, and external resistors. The base resistance is especially important because its thermal noise can be referred to the input of the transistor and amplified by

the circuit.

2.2. Shot Noise

Shot noise is caused by the discrete nature of electric charge. Current is not a perfectly smooth flow. Instead, it is made of many individual electrons or holes crossing a potential barrier. Because the arrival of these carriers has randomness, current noise is produced.

For a DC current I, the shot noise current spectral density is often written as: $S_i=2qI$, where q is the elementary charge.

In a BJT, shot noise is associated with junction currents, especially base current and collector current. The base-emitter junction behaves like a forward-biased diode, so carrier injection across this junction produces shot noise. Collector current can also contribute to shot noise because it is also made of discrete carrier transport.

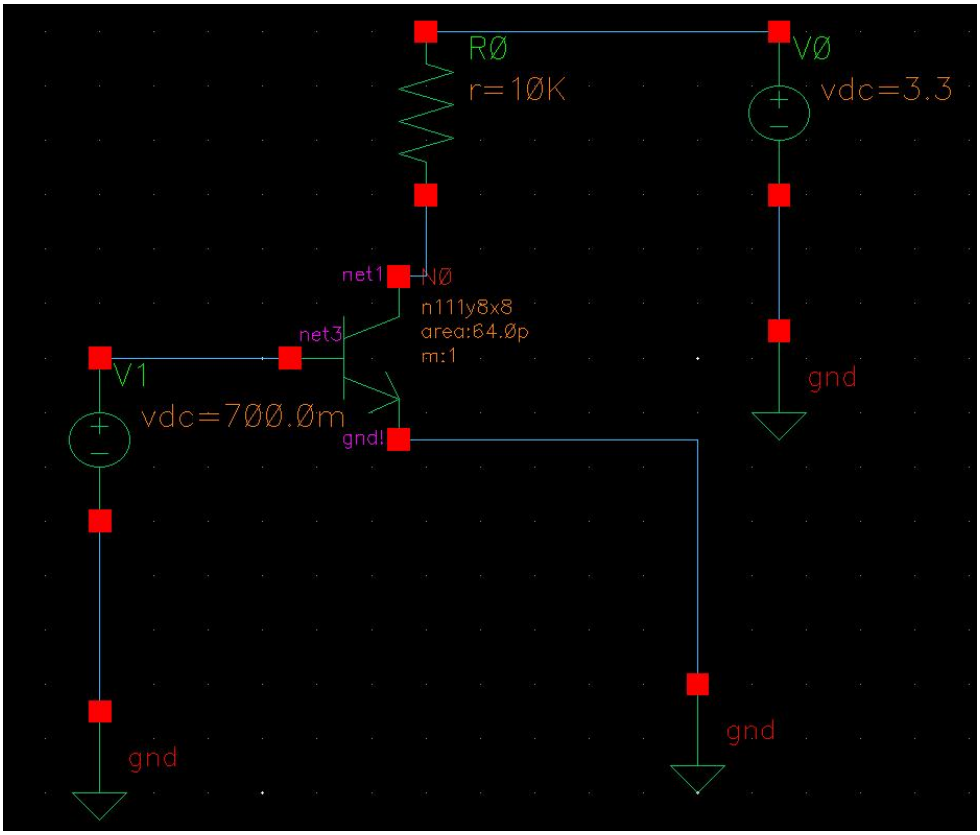
Shot noise is also white noise in its simplest form. This means that, unlike flicker noise, its spectral density does not increase strongly at low frequency.

2.3. Flicker Noise

Flicker noise is different from thermal noise and shot noise. It is strongest at low frequency and usually decreases as frequency increases. Its spectral density is often described by: $S(f)=K/f^{\gamma}$, where K is a noise coefficient and γ is close to 1.

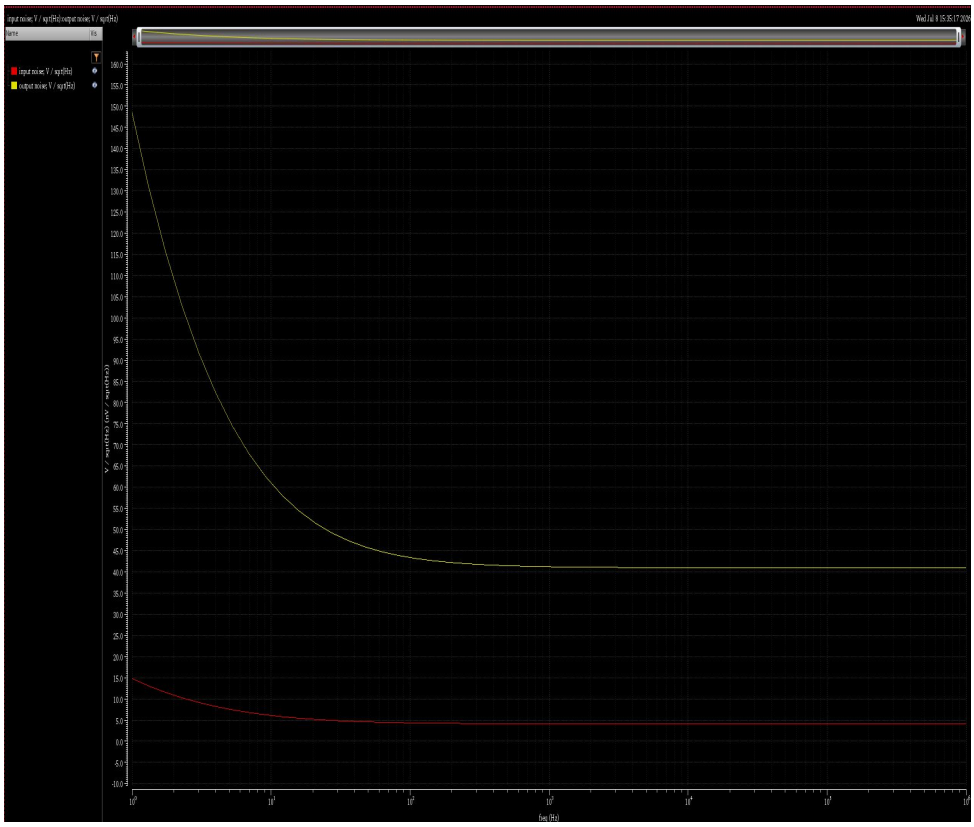
In BJTs, flicker noise is usually related to slow random changes inside the device. These changes can be caused by traps that capture and release carriers, defects that assist recombination, or imperfect surfaces and interfaces. Because these processes are slow compared with carrier transport, they strongly affect the low-frequency noise spectrum.

Circuit Schematic1:



Simulation1

The simulated output noise spectrum shows a clear low-frequency 1/f region and a high-frequency white-noise floor, which agrees with the compact flicker noise model.



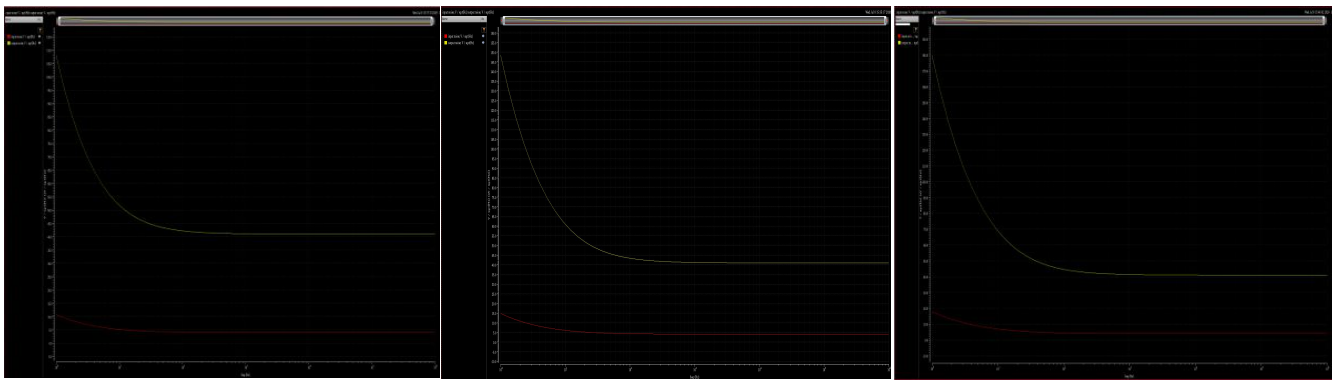
A common way to describe BJT flicker noise is to represent it as a base current noise source: $SI_B(f)=K_F I_B^{AF}/f^{FE}$

Here, S_{IB} is the base current noise spectral density, I_B is the DC base current, K_F is the flicker noise coefficient, A_F describes the current dependence, and F_E describes the frequency dependence.

Simulation2

To further investigate the effect of the flicker noise coefficient K_f on the 1/f noise performance of the BJT circuit, we run parametric simulations with three different K_f values: 0.964*10^{-15}, 1.964*10^{-15}, and 2.964*10^{-15}.

As shown in the simulation curves, the low-frequency noise spectral density rises monotonically as K_f increases, and the entire low-frequency noise segment shifts upward. In contrast, the three curves nearly overlap in the high-frequency white noise region, with no obvious difference. This result verifies that K_f is the dominant parameter controlling the magnitude of flicker noise, while it has negligible influence on the thermal noise and shot noise of the transistor.



K_f = 0.964*10^{-15}　　K_f = 1.964*10^{-15}　　K_f = 2.964*10^{-15}

It is well established in the literature that the flicker noise current exponent A_F takes a value close to 2 for most silicon bipolar junction transistors. This characteristic is consistently observed across mainstream process nodes and device architectures, and its physical origin is attributed to generation-recombination (g-r) noise arising from deep-level trap states within the depletion region of the emitter-base junction (Deen et al., 1999; Simoen et al., 1998).

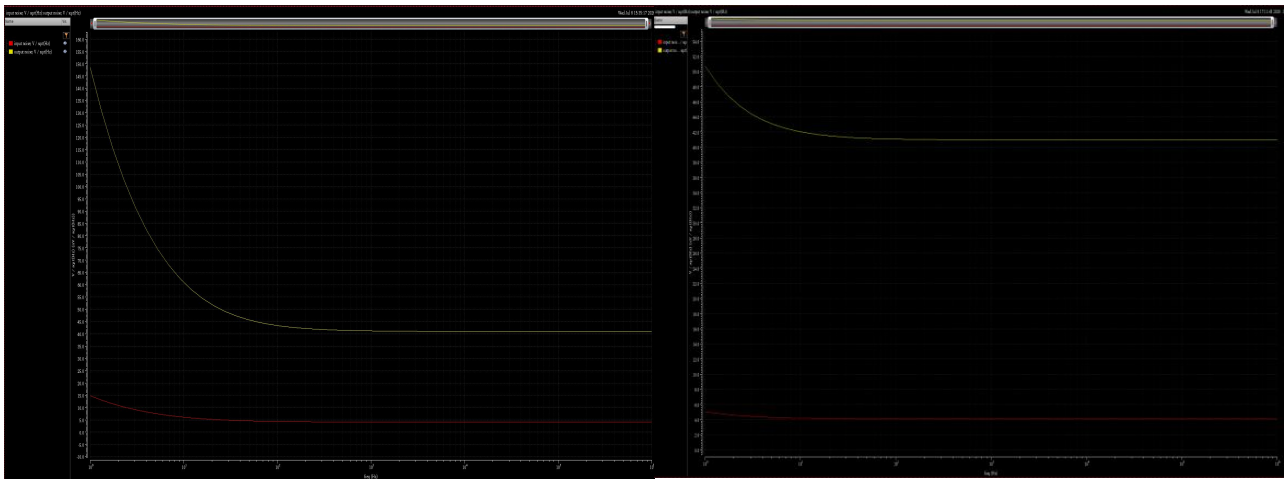
In this regime, the random trapping and detrapping of minority carriers by junction defects modulates the net recombination rate in the space-charge region. Since the recombination rate is proportional to the product of electron and hole concentrations, which scales with the square of the injected base current I_B, the resulting flicker noise spectral density exhibits a quadratic dependence on bias current. This gives rise to the canonical form of base-referred flicker noise: $S_{IB}(f) \propto I_B^2/f$

Simulation3

Following the analysis of the flicker noise coefficient K_F, the third set of simulations is conducted to verify the role of the flicker noise current exponent A_F in shaping the noise spectrum of the BJT circuit. All bias conditions, load parameters and the K_F coefficient remain constant throughout the test. Two representative values of A_F, 0.9162 and 1.1162, are selected for comparative analysis.

As shown in the simulation curves, a higher A_F leads to an overall reduction in noise spectral density across the low-frequency band. The 1/f noise segment shifts downward noticeably as A_F increases from 0.9162 to 1.1262. In contrast, the two curves converge and nearly overlap in the high-frequency white noise region, with no significant difference in thermal noise level.

This observation is fully consistent with the theoretical formulation of BJT flicker noise: $SI_B(f)=K_F I_B^{AF}/f^{FE}$ .Since the base bias current I_B of the device is on the microampere scale, the term $I_B\hat{}\{A_F\}$ decreases as the exponent A_F rises, directly suppressing the magnitude of flicker noise. As a result, the contribution of 1/f noise to the total spectrum weakens, making the overall noise characteristic approach flat white noise. It is also notable that the slope of the noise spectrum versus frequency remains unchanged for both A_F values, confirming that A_F only modulates noise amplitude through bias current scaling, and does not alter the frequency dependence of flicker noise.



$A_f = 0.9162$ $A_f = 1.1162$

When device geometry is considered, the noise spectral density is further inversely proportional to the emitter area A_E. This scaling law arises from the spatial averaging of uncorrelated trap fluctuations across a larger junction area, which suppresses the overall noise magnitude. The simplified area-normalized expression is written as: $S_{IB}(f) \propto I_B^2/A_E f$. Where A_E is emitter area.

3. Physical Origin of Flicker Noise

The physical origin of flicker noise in silicon bipolar junction transistors is closely related to the microscopic transport and recombination processes inside the device. Unlike thermal noise, originates from random thermal motion, and shot noise, which originates from the discreteness of charge transport, flicker noise is generally associated with slow fluctuations in material,

interface, and junction properties. These fluctuations modulate the terminal currents of the transistor and therefore appear as low-frequency current or voltage noise.

In silicon BJTs, the most widely accepted interpretation is that flicker noise is mainly associated with fluctuations of the base current. Since the base current contains recombination components in addition to ideal carrier injection, it is highly sensitive to traps, defects, and surface or interface states near the base-emitter junction. These localized defect states can randomly capture and emit carriers, modify recombination rates, and produce slow fluctuations in the base current. When many such microscopic fluctuations with a broad distribution of time constants are superimposed, the resulting spectrum can approach a 1/f-type behavior.

Therefore, BJT flicker noise is often discussed in terms of four related mechanisms: base current fluctuation, trap-assisted recombination, surface and interface defects, and Hooge-type mobility fluctuation. These mechanisms are not necessarily independent. In many practical silicon BJTs, traps and interface defects influence recombination processes, recombination processes modulate the base current, and the resulting terminal current fluctuation is modeled as an equivalent flicker noise source.

3.1. Base Current Fluctuation

A key feature of BJT flicker noise is its strong connection with base current fluctuation. In an ideal forward-active BJT, the collector current is mainly determined by minority-carrier injection from the emitter into the base, while the base current is associated with recombination and imperfect injection efficiency. In real devices, the base current contains several components, including recombination in the base-emitter depletion region, recombination at the emitter-base surface, and recombination related to interface states or defects.

Because of this physical composition, the base current is more sensitive to defects and surface conditions than the collector current. Small variations in recombination activity can produce noticeable variations in I_B These fluctuations can then be represented at the circuit level as an equivalent current noise source connected to the base terminal.

For many silicon BJTs, experimental results show that the current exponent A_F can be close to 2 under certain bias and geometry conditions. In that case, the flicker noise term becomes approximately: $SI_B(f) \propto I_B^2 f$.

This behavior also explains why bias selection is important in low-noise BJT circuit design. Increasing the bias current may improve transconductance and high-frequency performance, but it can simultaneously increase the low-frequency base current noise. Therefore, the optimum bias point is usually determined by a trade-off among gain, bandwidth, power consumption, and flicker noise.

Simulation4

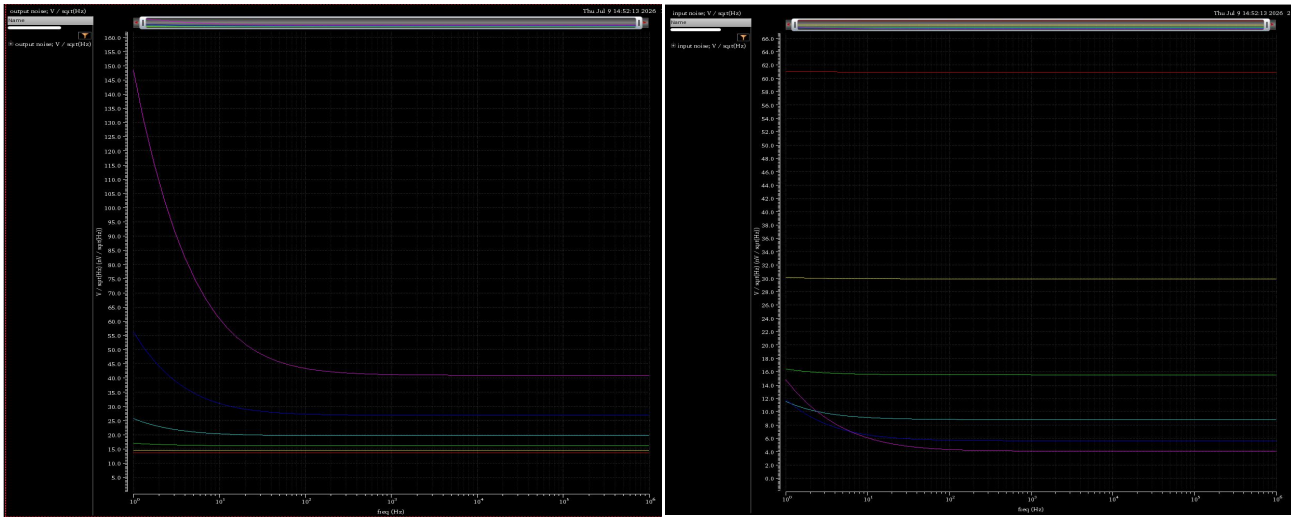
The base bias voltage determines the DC operating point of the BJT by regulating the forward bias state of the emitter-base junction. As the bias voltage Vbias increases, the forward bias of the emitter-base junction rises, causing the base current I_B to grow exponentially. Through the current amplification effect of the transistor, the collector current $I_C = \beta I_B$ increases synchronously. The bias dependence of both currents forms the physical basis for the evolution of device noise characteristics with operating points.

The output noise spectrum is shown in the left plot. The noise spectra under all bias conditions exhibit typical characteristics of BJT devices: the noise spectral density decreases monotonically with increasing frequency in the low-frequency band, which is dominated by 1/f flicker noise; in the high-frequency band, the noise amplitude remains constant, corresponding to the white noise floor composed of thermal noise and shot noise. As Vbias increases and both I_B and I_C rise, the overall amplitude of output noise increases significantly, and the growth in the low-frequency flicker noise region is markedly larger than that in the high-frequency white noise region. This phenomenon is fully consistent with the physical mechanism of flicker noise: on the one hand, the increase in base current intensifies the fluctuation of recombination carriers in the depletion region and on the device surface, enhancing the strength of the equivalent flicker current noise source at the base terminal; on the other hand, the rising collector current increases the transistor transconductance and the circuit voltage gain accordingly, further amplifying the noise.

The input-referred noise spectrum is shown in the right plot, and its variation trend is opposite to that of output noise: higher bias voltage and larger operating current lead to lower overall amplitude of input-referred noise. This phenomenon does not

contradict the aforementioned mechanism, but arises from the difference in the physical definition of the two noise parameters. Flicker noise is inherently a current noise source connected in parallel at the base terminal, and its magnitude increases with I_B; whereas input-referred noise is a voltage noise quantity referred to the input terminal, which is obtained by normalizing the output noise with the circuit voltage gain. As I_C increases, the transistor transconductance gm rises significantly and the input impedance r_π decreases accordingly. The effect of gain improvement outweighs the increase in base current noise, eventually causing the equivalent input voltage noise to decrease with rising bias current.

The above results intuitively verify the core trade-off in low-noise BJT design: increasing the bias current can improve circuit gain and optimize high-frequency input noise performance, but it also strengthens the base flicker noise source and elevates the low-frequency noise level at the output. Therefore, in circuit design for low-frequency sensing, audio amplification and other applications, the optimal operating point requires a comprehensive trade-off among gain, power consumption and flicker noise performance.



Red: Vbias=0.60V, Yellow: Vbias=0.62V, Green: Vbias=0.64V, Lightblue: Vbias=0.66V, Darkblue: Vbias=0.68V, Purple: Vbias=0.70V

3.2. Trap-Assisted Recombination

Trap-assisted recombination is another important physical source of BJT flicker noise. A trap is an energy state inside the semiconductor bandgap. It can capture an electron or a hole and later release it. These capture and release events are random.

In a BJT, traps can exist in the base-emitter depletion region, near the emitter-base junction, or near oxide-semiconductor interfaces. When traps participate in recombination, they can change the base current. If a trap captures a carrier, the local recombination condition may change. If it releases the carrier, the condition changes again. This produces random current fluctuation.

A single trap may produce random telegraph signal noise, also called RTS noise. In the time domain, RTS noise looks like the current randomly switching between two levels. In the frequency domain, one trap often produces a Lorentzian spectrum. When many traps with different time constants are present, the sum of many Lorentzian spectra can look like 1/f noise.

This idea is supported by experimental studies. Deen et al. observed RTS and Lorentzian spectra in small-area silicon BJTs. They also found that averaging the spectra of many small devices produced a 1/f-like spectrum similar to that of large-area devices (Deen et al., 1999). This is strong evidence that microscopic trapping and detrapping processes can form macroscopic flicker noise.

3.3. Surface and Interface Defects

Surface and interface defects are especially important in BJT flicker noise. A real transistor is not made of perfect materials. There are oxide layers, semiconductor interfaces, sidewalls, isolation regions, polysilicon contacts, and passivation layers. Defects near these regions can act as traps and recombination centers.

In polysilicon-emitter BJTs, a thin interfacial layer often exists between the polysilicon emitter and the crystalline silicon

emitter region. This layer can contain defects and trap states. Deen et al. argued that this interfacial region is a likely source of 1/f noise in polysilicon-emitter BJTs (Deen et al., 1999).

Simoen et al. also showed that BJT base current flicker noise can include both an area-related component and a perimeter-related component. The area-related component is associated with the emitter interface, while the perimeter-related component is associated with surface recombination near the emitter-base edge (Simoen et al., 1998).

A simplified form of this idea is: $SI_B(f)=K_aI_B^2/A_Ef^\gamma+K_pI_B^nP_Ef^\gamma$,where A_E is emitter area and P_E is emitter perimeter. The first term represents interface-area-related noise. The second term represents edge or perimeter-related noise.

3.4. Hooge-Type Mobility Fluctuation

Another possible explanation of 1/f noise is mobility fluctuation. Mobility describes how easily carriers move through a material. If the carrier mobility randomly fluctuates, the current also fluctuates.

A widely known empirical model is the Hooge relation: $S_I/I^2=\alpha_H/Nf$, where S_I is current noise spectral density, I is DC current, α_H is the Hooge parameter, and N is the number of free carriers.

This equation means that the normalized noise decreases when the number of carriers increases. It is often useful for bulk conduction noise in resistive materials and some semiconductor devices.

However, in silicon BJTs, flicker noise is often more directly related to base current fluctuation, recombination, and interface traps rather than pure bulk mobility fluctuation. Therefore, Hooge-type models can be useful conceptually, but compact BJT noise models more commonly use base current flicker noise expressions

4. Dependence of Flicker Noise on Device, Bias, Geometry, and Process

Flicker noise in silicon BJTs is not a constant value. It depends strongly on the transistor bias condition, device geometry, surface and interface quality, and fabrication process. This dependence is important because it helps us understand what causes flicker noise and how it can be reduced in practice.

In many silicon BJTs, flicker noise is closely related to base current fluctuation. A useful simplified relationship is: $SI_B(f)=K_FI_B^{AF}/A_Ef^{FE}$. This expression shows three basic trends: larger base current usually increases flicker noise; larger emitter area usually reduces normalized flicker noise; lower frequency gives stronger flicker noise.

However, the actual noise level also depends strongly on traps, recombination centers, and interface defects inside the device.

4.1. Bias Current Dependence

Bias current is one of the most important factors affecting flicker noise. Since BJT flicker noise is often modeled as base-current noise, increasing the base current usually increases the noise level ($SI_B(f)\propto I_B^{AF}/f^{FE}$).

Therefore, higher bias current does not always mean better noise performance. Although increasing collector current can improve transconductance and gain, it usually also increases base current, which may increase flicker noise.

For practical design, there is a trade-off: higher current can improve gain and bandwidth; higher current can also increase base-current flicker noise; the optimum bias point depends on the circuit application.

For example, in low-source-resistance amplifiers, a BJT can still be a good low-noise choice. But in high-source-resistance circuits, base-current noise can be converted into voltage noise through the source resistance

4.2. Geometry Dependence

Device geometry also strongly affects flicker noise. The most important geometry factors are emitter area and emitter perimeter.

For area-related flicker noise, a larger emitter area usually reduces the noise. The reason is that a larger area contains more microscopic noise sources. Their random fluctuations are averaged over a larger region, so the normalized noise becomes smaller.

This is why low-noise precision circuits often use larger BJT devices. However, larger devices also have larger capacitance, which may reduce speed and bandwidth. Therefore, increasing emitter area is useful for low-frequency precision circuits, but it may not be suitable for RF circuits.

Emitter perimeter is also important, especially for small transistors. Defects near the emitter-base edge can cause surface recombination and base-current fluctuation. Small devices usually have a larger perimeter-to-area ratio, so edge defects can become more important.

In simple terms: large emitter area helps reduce area-related flicker noise; small devices are more sensitive to edge defects; better edge passivation can reduce perimeter-related noise.

4.3. Surface and Interface Defects

Surface and interface defects are major physical causes of BJT flicker noise. These defects can act as traps or recombination centers. They can randomly capture and release carriers, causing base-current fluctuation.

Important defect regions include: the emitter-base junction; the oxide-silicon interface; the polysilicon-silicon interface; the emitter edge and sidewall regions; isolation and passivation interfaces.

In polysilicon-emitter BJTs, the interface between polysilicon and crystalline silicon can be an important flicker-noise source. If this interface contains many defects, the base-current noise can increase.

Therefore, interface quality is very important. A cleaner interface, fewer traps, and better passivation usually lead to lower flicker noise.

4.4. Process Dependence

Flicker noise is strongly affected by the fabrication process because processing steps can create defects and traps. Even if two BJTs have similar DC characteristics, they may have different flicker-noise levels if their interface quality is different.

Important process-related factors include: oxide quality; surface cleaning; plasma etching damage; ion implantation damage; annealing conditions; polysilicon deposition quality; passivation quality.

For example, plasma etching can damage sidewalls and create trap states. Ion implantation can create lattice defects. If annealing is not optimized, some defects may remain and increase recombination-related noise.

To reduce flicker noise, process improvements may include: improving surface cleaning before deposition or oxidation; reducing plasma damage; optimizing implantation and annealing; improving oxide and interface quality; improving polysilicon-emitter interface control; using hydrogen or forming-gas annealing to passivate dangling bonds; improving final passivation to protect the device surface. These improvements reduce trap density and recombination centers, which directly helps lower flicker noise.

4.5. Stress, Aging, and Radiation Effects

Flicker noise can also increase after electrical stress, aging, or radiation exposure. These effects can create new traps and interface states.

For example: hot-carrier stress can damage junction or interface regions; long-term aging can activate or generate additional defects; radiation can create oxide charge, interface traps, and lattice defects. These new defects increase carrier trapping and recombination fluctuation, leading to larger flicker noise.

Therefore, flicker noise can also be used as a reliability indicator. Sometimes a device may still show normal DC behavior, but its 1/f noise increases, indicating hidden degradation.

5. Circuit-Level Impact

Flicker noise matters because it appears in real circuits. Even if the noise originates from microscopic traps inside a transistor, the final effect may be seen as output voltage noise, reference voltage noise, or oscillator phase noise.

5.1. Low-Frequency Amplifiers

Low-frequency amplifiers are strongly affected by flicker noise because their signal band overlaps the low-frequency region where 1/f1/f noise is large. Examples include sensor interfaces, biomedical amplifiers, audio preamplifiers, and precision instrumentation amplifiers.

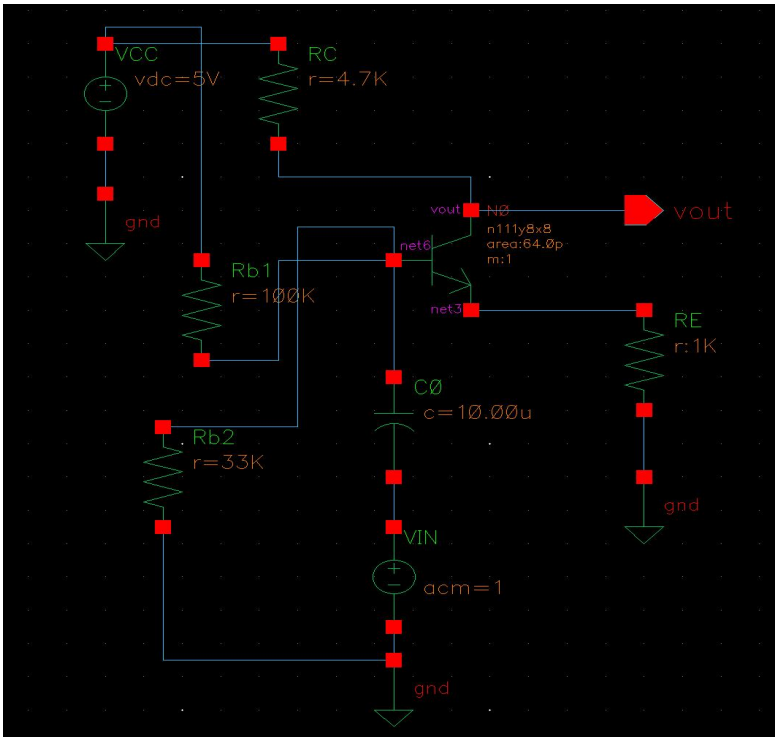
In a BJT amplifier, base current flicker noise can be converted into input voltage noise through source resistance or transistor

input impedance. It can then be amplified and appear at the output.

For example, if a base current noise i_{nB} flows through a source resistance R_S, it creates a voltage noise approximately equal to: $Vn=i_{nB}R_S$. This shows why source resistance is important. A high source resistance can convert even a small base current noise into a noticeable voltage noise.

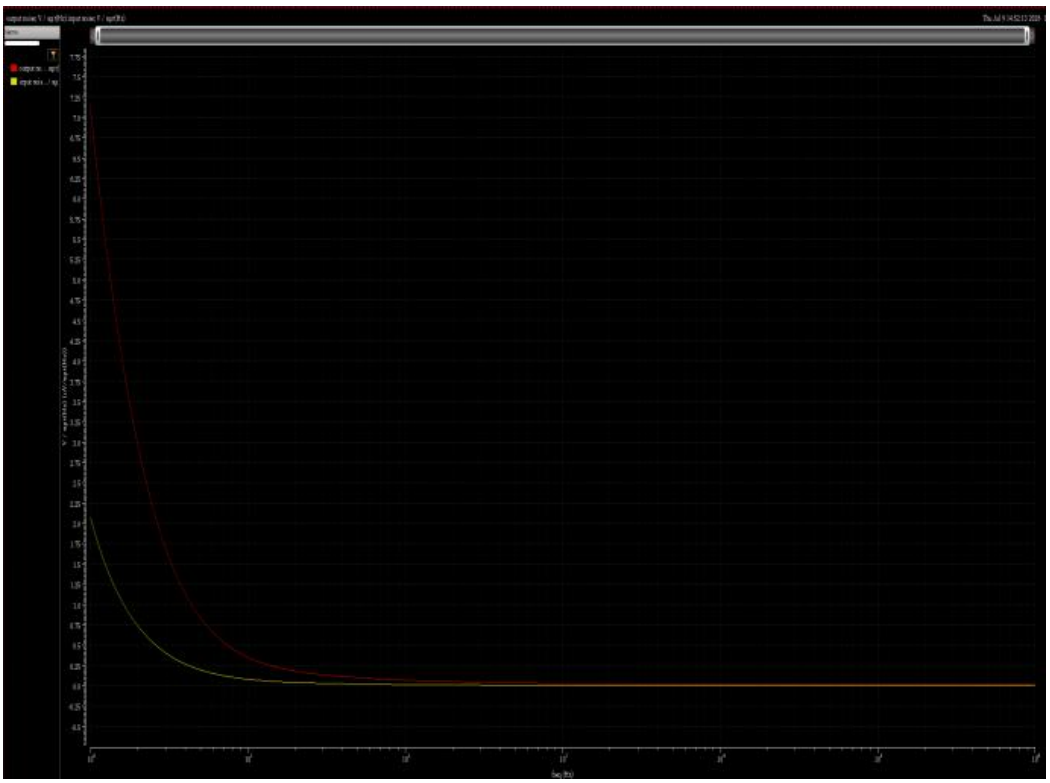
Therefore, for low-frequency BJT amplifier design, engineers must consider both voltage noise and current noise. A BJT may be a good choice for low source resistance, but for high source resistance, base current noise may become a serious limitation.

Circuit Schematic2:

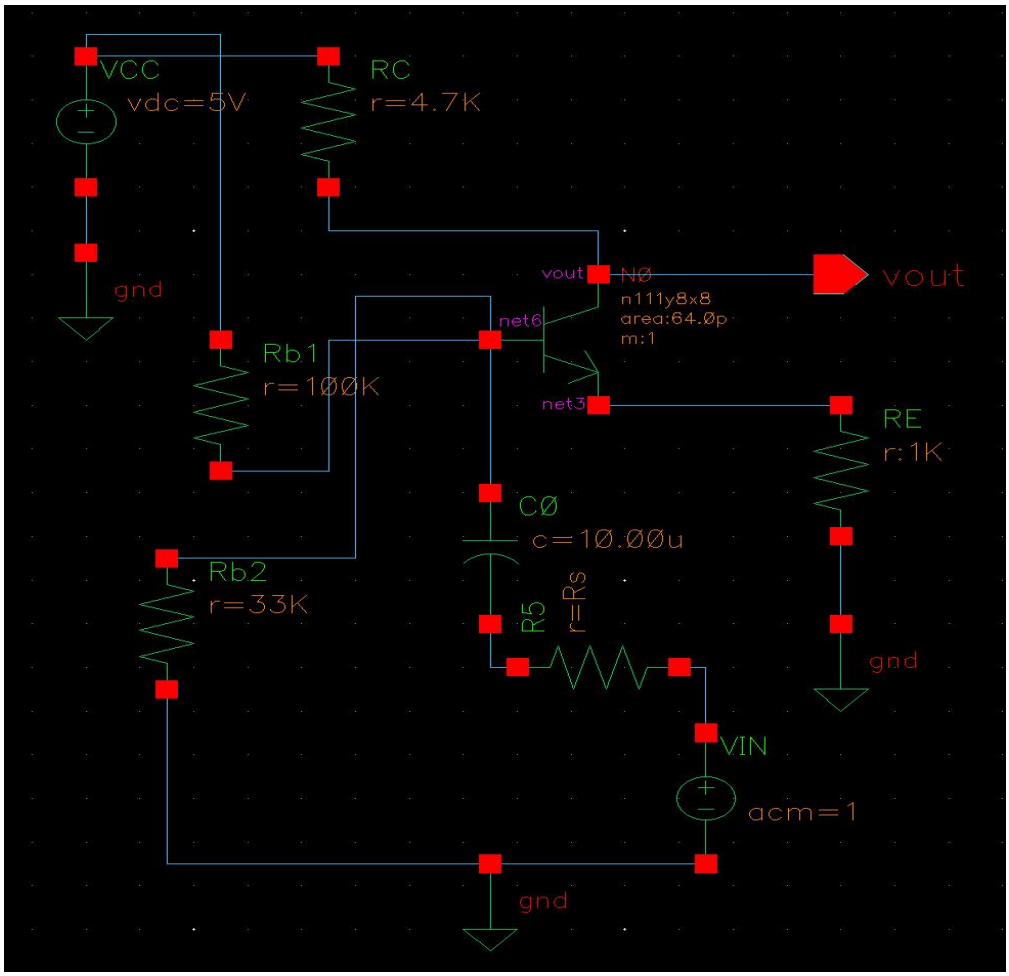


Simulation5:

To validate the foregoing theoretical analysis of flicker noise in low-frequency BJT amplifiers, a voltage-divider biased common-emitter BJT amplifier is designed and subjected to noise spectral density simulation. With a 5 V supply, the circuit adopts a 4.7 kΩ collector resistor, a 1 kΩ emitter degeneration resistor, bias resistors of 100 kΩ and 33 kΩ, and a 10μF input coupling capacitor; noise characterization is conducted over a frequency range from 1 Hz to 1 MHz, with both output and input-referred voltage noise spectral density extracted for analysis. The simulation results present a typical 1/f noise profile in the low-frequency band below roughly 100 Hz, where both output and input-referred noise increase significantly with decreasing frequency, directly confirming that flicker noise dominates the total noise of BJT amplifiers in the low-frequency region. As frequency rises above 100 Hz, the noise curves gradually level off to a white noise plateau, meaning thermal noise from the transistor and resistors becomes the primary noise source while the contribution of flicker noise fades rapidly. In addition, the output noise spectral density is consistently higher than the input-referred value, which manifests that the voltage gain of the amplifier amplifies input noise synchronously with the signal. The results also corroborate the impact of source resistance on noise performance: with an ideal near-zero-resistance input source adopted in this simulation, the low-frequency flicker noise is mainly derived from the base current noise converted through the transistor's input impedance and equivalent bias resistance, and in line with the relation $Vn = i_{nB}R_S$, a higher source resistance would convert the same base current noise into larger input voltage noise and impose a more severe limitation on noise performance.



Circuit Schematic3:

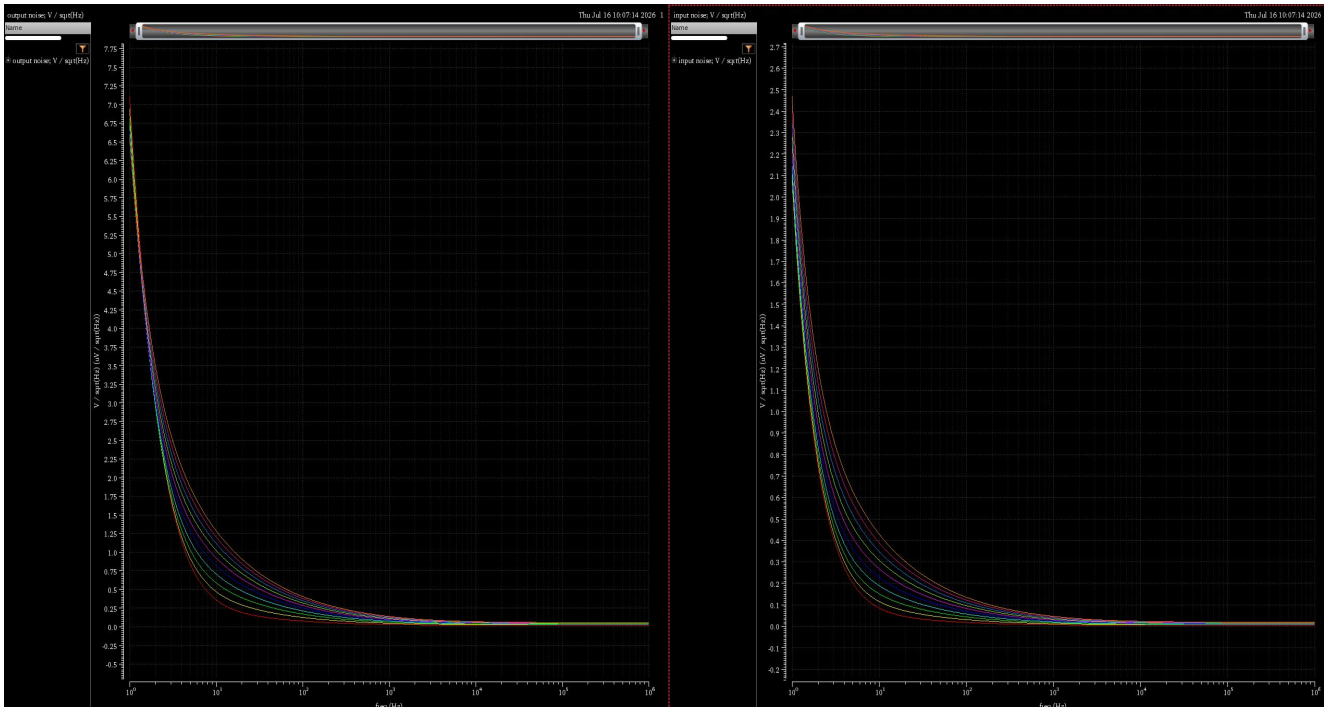


Simulation6:

To quantitatively evaluate the circuit-level conversion of base-current flicker noise into input-referred voltage noise, a parametric sweep of source resistance was conducted on a voltage-divider biased common-emitter BJT amplifier. The source resistance RS was varied from 10 Ω to 10 kΩ, while all DC bias conditions, load resistances, and transistor device parameters were kept constant. Both output-referred and input-referred noise spectral densities were simulated across a frequency range of 1 Hz to 1 MHz.

The simulated input-referred noise spectra show a strong dependence on source resistance, with distinct behaviors in two frequency regimes. In the low-frequency region (below approximately 100 Hz) where flicker (1/f) noise dominates, the noise magnitude rises significantly with increasing source resistance. This observation confirms that base-current flicker noise is converted into equivalent input voltage noise through the source impedance. In the high-frequency white-noise region (above approximately 1 kHz), the noise floor also increases monotonically with RS ; this elevation stems primarily from the thermal noise of the source resistor and base-current shot noise, and is unrelated to flicker noise mechanisms.

These findings verify that source resistance is a critical design parameter for low-frequency BJT amplifiers. Deliberate optimization of source impedance and bias network resistance is necessary to suppress the contribution of base-current flicker noise to the overall input-referred noise performance.



5.2. Bandgap References

Bandgap references are circuits that generate a stable reference voltage. They are widely used in analog and mixed-signal systems. A typical bandgap reference uses BJT base-emitter voltages and thermal voltage differences to create a voltage that is relatively stable with temperature.

Flicker noise in the BJTs of a bandgap reference can directly contribute to reference voltage noise. Because bandgap references often operate at low frequency or DC, flicker noise can be especially important.

Noise in a bandgap reference is important because many circuits depend on the reference voltage. If the reference is noisy, analog-to-digital converters, bias circuits, sensors, and precision amplifiers may also suffer.

To reduce flicker noise in bandgap references, designers may use larger emitter-area BJTs, careful bias selection, filtering capacitors, chopping techniques, or averaging of multiple devices. The key idea is to reduce device-level noise and prevent it from appearing at the final reference output.

5.3. RF Oscillators and Phase Noise

Although flicker noise is a low-frequency noise source, it can affect RF oscillators. This happens because low-frequency noise can be upconverted around the carrier frequency. In an oscillator, this appears as close-in phase noise.

A noisy oscillator output can be written conceptually as: $v(t)=A(t)\cos(2\pi f_0 t+\phi(t))$, where A(t) is amplitude fluctuation, f0 is carrier frequency, and $\phi(t)$ is phase fluctuation.

Flicker noise inside the active transistor can modulate the oscillator current, capacitance, or timing. These fluctuations can become phase fluctuations. Walls et al. explained that 1/f noise in BJT amplifiers can contribute to both amplitude modulation noise and phase modulation noise (Walls et al., 1995).

This is especially important near the carrier. In many oscillators, close-in phase noise has a strong contribution from device flicker noise. Therefore, reducing BJT flicker noise can improve oscillator spectral purity.

6. Practical Techniques for Reducing Flicker Noise in Silicon BJTs

The reduction of flicker noise in silicon BJTs requires control of the physical regions where base-current fluctuations are generated. Since the dominant contribution is usually associated with carrier trapping, detrapping, and trap-assisted recombination near the emitter-base junction and related interfaces, effective reduction techniques should aim to decrease the density of electrically active defects, suppress surface recombination, improve current uniformity, and reduce the circuit-level conversion of base-current noise into voltage noise. In practice, the most important approaches are process-level interface improvement, reduction of plasma and implantation damage, defect passivation by annealing, geometry and layout optimization, and careful bias and circuit design.

6.1. Improvement of Interface Quality and Emitter-Base Junction Cleanliness

A major practical method for reducing flicker noise is to improve the cleanliness and structural quality of the interfaces involved in carrier injection and recombination. In silicon BJTs, the emitter-base junction and nearby oxide-silicon or polysilicon-silicon interfaces strongly influence the base current. If these regions contain contamination, dangling bonds, native oxide residues, or nonuniform interfacial layers, they can introduce localized trap states inside the bandgap. These traps randomly capture and emit carriers and modulate the recombination component of the base current, which appears as low-frequency current fluctuation.

Improved interface quality reduces flicker noise because it lowers the number of active traps that can participate in recombination. Before oxidation, polysilicon deposition, or emitter formation, wafer cleaning must remove organic residues, metallic impurities, particles, and uncontrolled native oxide. For polysilicon-emitter BJTs, this step is especially important because the thin interfacial layer between polysilicon and crystalline silicon can strongly affect carrier transport. A poorly controlled interfacial oxide can create spatially nonuniform injection and additional recombination centers, increasing base-current flicker noise. By using optimized surface cleaning and controlled interfacial oxide formation, the emitter current becomes more uniform and the density of recombination-active defects is reduced.

This technique does not simply reduce noise by changing the DC current level. Instead, it changes the microscopic quality of the junction and interface. For the same bias current, a cleaner and more uniform interface produces fewer random recombination events, so the base-current noise spectral density decreases. Therefore, interface engineering is one of the most fundamental process-level methods for reducing BJT flicker noise.

6.2. Reduction of Plasma-Etching and Ion-Implantation Damage

Another important method is to reduce process-induced damage caused by plasma etching and ion implantation. These steps are necessary in modern bipolar and BiCMOS fabrication, but they can create electrically active defects near critical junctions and sidewalls. Plasma etching may damage silicon surfaces through ion bombardment, while ion implantation can displace atoms from their lattice sites and generate residual point defects or defect clusters. If these defects remain near the emitter-base depletion region, emitter perimeter, or isolation sidewalls, they can act as trap-assisted recombination centers.

The connection between process damage and flicker noise is direct. Flicker noise in BJTs is strongly related to base-current fluctuation, and the base current is sensitive to recombination at defects. When plasma damage or implantation damage creates additional trap states, the local recombination rate becomes more unstable. The random occupation of these trap states produces

fluctuations in recombination current, increasing the low-frequency base-current noise.

To reduce this effect, plasma processes should be optimized to minimize ion bombardment near sensitive junctions, and post-etch cleaning or recovery annealing should be used to remove or repair damaged surface layers. Similarly, implantation dose, energy, and annealing conditions should be chosen so that dopants are activated while residual lattice damage is reduced. Proper annealing is especially important because it can repair implantation-induced defects and restore crystalline order. However, the thermal budget must be controlled carefully, since excessive annealing can broaden junctions and degrade high-frequency device performance.

Thus, the purpose of damage reduction is not only to improve DC characteristics, but also to suppress the microscopic defect states that generate low-frequency recombination fluctuations. A device with lower residual process damage will generally show lower flicker noise, especially in small-area transistors where sidewall and perimeter-related defects can dominate.

6.3. Hydrogen Passivation and Forming-Gas Annealing

Hydrogen passivation is a widely used technique for reducing electrically active defects at silicon interfaces. Many interface defects are associated with dangling bonds, which introduce energy states in the silicon bandgap. These states can trap carriers and participate in recombination. During hydrogen or forming-gas annealing, hydrogen atoms diffuse to the interface and bond with dangling bonds, making them electrically inactive. As a result, the density of interface traps is reduced.

This process can reduce BJT flicker noise because it decreases the number of traps available for carrier capture and emission. When fewer traps are active near the emitter-base junction or sidewall regions, the recombination component of the base current becomes more stable. Since the flicker noise source is often modeled as a base-current noise source, reducing the fluctuation of this current directly lowers the low-frequency noise spectral density.

Forming-gas annealing is especially useful after oxidation, etching, or passivation deposition, because these steps can leave interface states or dangling bonds. In BJTs with significant perimeter-related noise, hydrogen passivation can reduce surface recombination near the emitter edge. In polysilicon-emitter structures, it can also improve the electrical quality of interfaces associated with the emitter contact and passivation layers.

However, hydrogen passivation must be considered together with reliability. Hydrogen can be redistributed or depassivated under later thermal processing or electrical stress. If passivated bonds are broken during long-term operation, traps may become active again and flicker noise may increase. Therefore, hydrogen annealing is effective for noise reduction, but its thermal stability and long-term reliability must be included in the process design.

6.4. Geometry and Layout Optimization

Device geometry has a strong influence on flicker noise because microscopic noise sources are distributed over the active area and along the device perimeter. For area-related flicker noise, increasing the emitter area reduces normalized noise by statistical averaging. A larger emitter contains more independent microscopic fluctuation sources, and their uncorrelated contributions average out more effectively. Therefore, for the same current density, a larger emitter area can reduce the normalized base-current flicker noise.

However, geometry optimization is not only a matter of increasing area. In small BJTs, perimeter-related noise can become significant because defects near the emitter edge and sidewall contribute to surface recombination. If the perimeter-to-area ratio is large, edge defects represent a larger fraction of the total recombination current, and the device becomes more sensitive to sidewall damage and passivation quality. Therefore, layout should be designed to reduce unnecessary perimeter contribution and avoid geometries that create strong edge-dominated current flow.

Uniform current distribution is also important. If current crowding occurs, a small region of the emitter carries a disproportionately large current density. In that case, local defects in this region have a stronger effect on total base-current fluctuation. By designing emitter geometry, contacts, and layout symmetry to promote uniform injection, the influence of individual local traps can be reduced.

The trade-off is that larger or more carefully optimized devices usually introduce larger junction capacitance and occupy more chip area. This can reduce speed and bandwidth, which is undesirable in high-frequency applications. Therefore, geometry optimization is especially effective in low-frequency precision circuits, bandgap references, and sensor interfaces,

where low flicker noise is more important than maximum frequency performance.

6.5. Bias Optimization and Circuit-Level Suppression

Even when the device process is fixed, flicker noise can be reduced at the circuit level by choosing an appropriate operating point and suppressing the conversion of current noise into voltage noise. In BJTs, flicker noise is often represented as a base-current noise source whose magnitude increases with base current.

This means that increasing bias current can increase the intrinsic base-current flicker noise. At the same time, a higher collector current increases the transconductance of the transistor: $g_m=I_C/V_T$, which can reduce input-referred thermal voltage noise and improve gain. Therefore, the optimum bias point is determined by a balance between increased transconductance and increased base-current noise.

The source resistance is especially important in this trade-off. Base-current noise flowing through the source resistance produces an equivalent input voltage noise: $v_n=i_{nB}RS$.

For low-source-resistance circuits, this conversion is weak, and a BJT can provide excellent low-noise performance. For high-source-resistance circuits, even a small base-current flicker noise can generate significant input voltage noise, making the BJT less favorable unless the bias and device size are carefully chosen.

Circuit techniques can further suppress the practical impact of flicker noise. In precision amplifiers, chopping can move the signal to a higher frequency band where flicker noise is lower and then demodulate it back to baseband. Auto-zeroing and correlated double sampling reduce low-frequency noise by sampling and subtracting slowly varying offset and noise components. In bandgap references or bias circuits, low-pass filtering and device averaging can reduce the noise appearing at the final output. These methods do not remove the physical traps inside the transistor, but they reduce how much device-level flicker noise affects the system-level signal.

In summary, bias and circuit optimization are necessary because the lowest device noise does not always correspond to the best circuit noise performance. The optimum design depends on source resistance, bandwidth, gain requirement, power consumption, and the relative importance of low-frequency noise.

7. Conclusion

This paper has reviewed the physical origin, compact modeling, and circuit-level impact of flicker noise in silicon bipolar junction transistors. Flicker noise is an important low-frequency noise mechanism in BJTs and is mainly associated with base current fluctuations caused by carrier trapping, detrapping, trap-assisted recombination, and surface or interface defects near the emitter-base junction. Compared with thermal noise and shot noise, flicker noise shows a strong frequency dependence and becomes dominant at low frequencies.

The compact modeling of BJT flicker noise is commonly based on an equivalent base current noise source, where the noise spectral density depends on the flicker noise coefficient, base current, current exponent, frequency exponent, and device geometry. Simulation results show that increasing the flicker noise coefficient raises the low-frequency noise level, while the high-frequency white noise floor remains almost unchanged. Bias current, emitter area, and source resistance also have significant influence on the observed noise performance.

At the circuit level, flicker noise can degrade the performance of low-frequency amplifiers, bandgap references, precision analog circuits, and RF oscillators. In amplifier applications, base current flicker noise can be converted into input voltage noise through source resistance and then amplified at the output. Therefore, low-noise BJT design requires careful optimization of device geometry, bias point, source impedance, layout, and process quality.

Overall, reducing flicker noise in silicon BJTs requires both device-level and circuit-level approaches. Improving interface quality, reducing process-induced defects, using larger emitter areas, optimizing bias current, and applying circuit techniques such as filtering or chopping can effectively reduce the impact of flicker noise. Understanding these mechanisms is essential for designing high-performance low-frequency and precision analog circuits.

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